Spconform: Spatially Localized Conformal Prediction and Spatial Diagnostic Auditing in R

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Abstract Quantifying predictive uncertainty for spatial data is critical across environmental monitoring, spatial epidemiology, and geosciences. Standard geostatistical interpolation methods such as Gaussian kriging rely on correctly specified covariance structures and stationarity, suffering substantial coverage breakdown under model misspecification. Although conformal prediction guarantees distribution-free, finite-sample predictive validity, its foundational assumption of exchangeability is systematically violated in the presence of spatial autocorrelation and heteroskedasticity. The spconform package resolves this foundational challenge on CRAN by providing distribution-free, model-agnostic, localized conformal calibration for geostatistical, areal lattice, count, multiclass classification, and multivariate spatial data. Built in Base R with zero heavy external dependencies, spconform introduces adaptive directional anisotropy rotation, five advanced spatial diagnostic audits, and demonstrates superior local coverage and interval efficiency compared to non-spatial conformal tools and Python's MAPIE library. This article presents the methodological foundations, software architecture, cross-system benchmarks, and reproducible case studies.

1 Introduction

Quantifying predictive uncertainty in spatial data is essential across diverse scientific disciplines, including soil contamination mapping, air quality surveillance, precision agriculture, and spatial epidemiology. When decisions involve environmental risk thresholds, public health interventions, or mineral resource estimation, point predictions alone are inadequate; practitioners require reliable prediction regions with guaranteed probabilistic coverage (Cressie, 1993; Bivand et al., 2013).

Historically, geostatistical interpolation has relied on Gaussian random fields and kriging. While kriging provides optimal linear unbiased predictions under the true data-generating process, its validity rests entirely on the correct parametric specification of the spatial covariance function (semivariogram) and Gaussian errors. In observational geographic studies, spatial fields frequently exhibit non-stationarity, local heteroskedasticity, directional anisotropy, and heavy-tailed disturbances. When these assumptions are violated, classical kriging prediction intervals suffer severe empirical coverage degradation, frequently dropping far below nominal confidence targets.

Conformal prediction, pioneered by Vovk et al. (2005) and advanced by Lei et al. (2018), offers an appealing alternative by constructing prediction intervals with mathematically guaranteed, finite-sample marginal coverage without requiring distributional or parametric modeling assumptions. However, standard conformal calibration theorems fundamentally rely on the exchangeability (or i.i.d. assumption) of the calibration and query observations. In spatial statistics, Tobler's First Law dictates that near observations are systematically more related than distant ones, introducing spatial autocorrelation and spatial covariate shifts (Tibshirani et al., 2019; Barber et al., 2023). Applying standard non-spatial conformal methods (such as those implemented in global split-conformal routines) produces spatially miscalibrated intervals: regions with high local variance suffer chronic undercoverage, while low-variance regions receive unnecessarily wide, inefficient intervals.

The **spconform** package (Jabbar, 2026) introduces a comprehensive, model-agnostic framework for spatial and spatio-temporal conformal prediction in the R environment. Published on the Comprehensive R Archive Network (CRAN) under the GPL-3 license, **spconform** provides localized conformal calibration using spatial proximity kernels and graph shortest-path network weights, adapting the theoretical foundations of localized non-exchangeable conformal inference (Barber et al., 2023; Mao et al., 2024).

This paper introduces the methodological principles, package architecture, multi-modality features, cross-system benchmarks against Python's state-of-the-art **MAPIE** library (Taquet et al., 2022), and applied case studies included in **spconform**.

2 Methodological framework

Localized split conformal prediction

Let $\{(s_i, Y(s_i))\}_{i=1}^n$ denote observed spatial data, where $s_i \in \mathcal{D} \subset \mathbb{R}^2$ represents geographic coordinates and $Y(s_i) \in \mathbb{R}$ is the observed continuous response. Given a query location $s_0 \in \mathcal{D}$ and a nominal miscoverage level $\alpha \in (0,1)$, our objective is to construct a predictive interval $\mathcal{C}_\alpha(s_0) = [\hat{l}(s_0), \hat{u}(s_0)]$ guaranteeing:

$$\mathbb{P}\left(Y(s_0) \in \mathcal{C}_\alpha(s_0) \mid S = s_0\right) \geq 1 - \alpha.$$

In split conformal prediction, the data indices $\{1, \ldots, n\}$ are partitioned into a training set $\mathcal{I}_{\text{train}}$ and a calibration set $\mathcal{I}_{\text{cal}}$. An arbitrary point prediction model $\hat{\mu}(\cdot)$ is trained strictly on $\mathcal{I}_{\text{train}}$. For each calibration location $s_i \in \mathcal{I}_{\text{cal}}$, we compute the absolute nonconformity residual:

$$R_i = |Y(s_i) - \hat{\mu}(s_i)|.$$

Under classical conformal inference, the empirical $(1-\alpha)$-quantile of $\{R_i\}$ is applied uniformly across the entire domain. In **spconform**, calibration residuals are localized via continuous spatial distance kernels $K_h(s_0, s_i)$. For a bandwidth $h > 0$, normalized spatial weights are assigned:

$$w_i(s_0) = \frac{K_h(\|s_0 - s_i\|)}{\sum_{j \in \mathcal{I}_{\text{cal}}} K_h(\|s_0 - s_j\|)}.$$

The localized prediction interval is then constructed as:

$$\mathcal{C}_\alpha(s_0) = \left[\hat{\mu}(s_0) - \hat{Q}_{1-\alpha}(s_0),\ \hat{\mu}(s_0) + \hat{Q}_{1-\alpha}(s_0)\right],$$

where $\hat{Q}_{1-\alpha}(s_0)$ is the $(1-\alpha)$-quantile of the spatially weighted empirical distribution:

$$F_{s_0}(r) = \sum_{i \in \mathcal{I}_{\text{cal}}} w_i(s_0)\, \mathbf{1}(R_i \leq r) + w_0(s_0)\, \mathbf{1}(\infty \leq r).$$

As established by Mao et al. (2024) and Barber et al. (2023), under local Hölder continuity of the conditional residual distribution, the local coverage gap decays at the rate $\mathcal{O}(h^\beta)$, ensuring asymptotic conditional validity and exact finite-sample robustness.

Automatic directional anisotropy rotation

Spatial continuity frequently exhibits directional asymmetry (anisotropy) due to geological faults, river corridors, or prevailing wind directions. When isotropy is incorrectly assumed, circular kernels over-smooth across the barrier and under-smooth along the continuity corridor.

spconform incorporates an automated directional coordinate rotator (`auto_anisotropy = TRUE`). The algorithm evaluates directional semivariances $\hat{\gamma}(\theta, h)$ across angle bins $\theta \in [0, \pi)$ via automated ANOVA F-testing. Upon detecting significant geometric anisotropy with ratio $\lambda = a_{\max}/a_{\min} > 1$ along principal orientation θ^*, spatial coordinates are transformed prior to kernel weighting:

$$\tilde{s} = s\, \mathbf{R}(\theta^*)\, \mathbf{S}(\lambda), \quad \mathbf{R}(\theta^*) = \begin{pmatrix} \cos\theta^* & -\sin\theta^* \\ \sin\theta^* & \cos\theta^* \end{pmatrix}, \quad \mathbf{S}(\lambda) = \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix}.$$

This transformation stretches distances perpendicular to the principal axis, ensuring that localized calibration borrowing strictly follows the physical continuity of the underlying process.

3 Multi-modality framework of spconform

While geostatistical continuous fields form the foundation of spatial analysis, applied researchers regularly encounter diverse data modalities. Version 0.2.2 of **spconform** expands localized conformal inference across five distinct modalities:

1. **Geostatistical Continuous Fields (`scp_geostatistical()`):** Continuous kernel-weighted intervals for point-referenced spatial and spatio-temporal coordinates, featuring automatic bandwidth selection, nearest-neighbor localized subsampling, and directional anisotropy rotation.
2. **Areal Lattice Networks (`scp_areal()`):** Graph-based leave-one-out conformal calibration over spatial contiguity matrices (e.g., polygon shapefiles, county borders), decaying calibration influence exponentially across graph topological shortest paths $d_G(i, j)$.

3. **Discrete Count and Epidemiology (`scp_count()`):** Spatially localized integer conformal prediction intervals for Poisson- and negative binomial-distributed disease counts, incorporating Pearson overdispersion adjustments, Anscombe variance-stabilizing transformations, and guaranteed non-negative integer bounds ($l \geq 0$).
4. **Spatial Multiclass Classification (`scp_classification()`):** Adaptive Prediction Sets (APS) and Least-Likely (LAC) conformal classification sets (Romano et al., 2020) adapted with spatial kernel calibration, outputting rigorous prediction sets $\hat{C}(s_0) \subseteq \{1, \ldots, K\}$ satisfying spatial coverage guarantees.
5. **Multivariate Vector Responses (`scp_multivariate()`):** Simultaneous joint prediction ellipsoids and projected coordinate-wise bounding boxes for vector spatial fields $Y(s) \in \mathbb{R}^d$ under localized Mahalanobis distance nonconformity scores:

$$R_i(s_0) = \sqrt{(Y(s_i) - \hat{\mu}(s_i))^\top \hat{\boldsymbol{\Sigma}}(s_0)^{-1} (Y(s_i) - \hat{\mu}(s_i))}.$$

4 Software architecture and interface

spconform is deliberately engineered following the Unix philosophy and CRAN guidelines:

- **Zero Heavy Dependencies:** The core package relies exclusively on standard Base R packages (**stats**, **graphics**, **grDevices**, **utils**), avoiding compilation vulnerabilities associated with external C++ geospatial libraries (GDAL, GEOS, PROJ).
- **Strict Model-Agnosticism:** Any point prediction model in R can be supplied via a unified functional wrapper:

```
pred_fun = function(s_train, y_train, s_new) { ... }
```

This design enables users to seamlessly wrap linear models, generalized additive models (**mgcv**), random forests (**ranger**), gradient boosting machines, or deep neural networks without modifying package internals. - **Unified S3 Object-Oriented Interface:** Standard R generics are consistently implemented across all modalities, including `print()`, `summary()`, `plot()`, `predict()`, `residuals()`, `diagnose()`, and `coverage_report()`.

5 Empirical benchmark and cross-system comparison

To evaluate the empirical performance of **spconform**, we conduct a controlled benchmark comparing four distinct spatial uncertainty paradigms: 1. **Classical Gaussian Kriging** (parametric geostatistical model with exponential variogram). 2. **Standard Global Conformal Prediction** (unweighted split conformal inference assuming exchangeability). 3. **Python MAPIE 1.6.0** (the world's most widely adopted open-source conformal library, executing non-spatial conformal calibration). 4. **spconform** (spatially localized conformal calibration).

Synthetic heteroskedastic spatial experiment

We simulate a synthetic spatial domain of $N = 250$ locations ($n_{\text{train}} = 175$, $n_{\text{test}} = 75$) exhibiting pronounced spatial heteroskedasticity and spatial autocorrelation:

$$Y(s) = \sin(s_1) + \cos(s_2) + \varepsilon(s), \quad \varepsilon(s) \sim \mathcal{N}\left(0, \sigma^2(s)\right),$$

where the localized noise standard deviation varies five-fold across the geographic domain: $\sigma(s) = 0.2 + 1.0 \times \mathbf{1}(s_1 + s_2 > 1.0)$. Nominal coverage is set to $1 - \alpha = 0.90$.

```
set.seed(42)
n <- 250
coords <- matrix(runif(n * 2, 0, 2), n, 2)
# Spatially varying heteroskedastic noise
sigma_s <- ifelse(coords[, 1] + coords[, 2] > 1.0, 1.2, 0.2)
y <- sin(coords[, 1]) + cos(coords[, 2]) + rnorm(n, sd = sigma_s)

idx_train <- 1:175
idx_test  <- 176:250
```

Table 1: Comparative empirical benchmark on heteroskedastic spatial simulation ($N = 250$, target $1 - \alpha = 90\%$).

Method	Target	Coverage	Mean Width	Winkler Score	Local Adaptivity
Classical Kriging	90.0%	73.3%	1.84	4.89	Poor (misspecified)
Global Conformal	90.0%	89.3%	2.51	3.65	Rigid (constant width)
Python MAPIE 1.6.0	90.0%	89.3%	2.51	3.65	Rigid (non-spatial)
spconform	90.0%	90.7%	2.08	2.94	Optimal (spatially adaptive)

```
s_tr <- coords[idx_train, ]; y_tr <- y[idx_train]
s_te <- coords[idx_test,  ]; y_te <- y[idx_test]

# Simple linear trend predictor function
pfun <- function(st, yt, sn) {
  fit <- lm(yt ~ st[, 1] + st[, 2])
  cbind(1, sn[, 1], sn[, 2]) %*% coef(fit)
}

# Fit localized conformal prediction
fit_sp <- scp_geostatistical(s_tr, y_tr, s_te, pfun, alpha = 0.10, seed = 42)
rep_sp <- coverage_report(fit_sp, y_te)
```

The benchmark results comparing empirical coverage, mean interval width, and the strictly proper Winkler Interval Score (WIS) (Winkler, 1972) are summarized in Table 1.

As demonstrated in Table 1, classical kriging suffers a catastrophic coverage collapse (73.3%) due to its rigid global variance assumption. Both global conformal prediction and Python's MAPIE 1.6.0 achieve marginal coverage (89.3%), but do so by inflating interval widths globally (mean width 2.51), failing to capture the five-fold spatial variance transition. In contrast, **spconform** accurately achieves the nominal 90.7% coverage with an 17.1% reduction in mean interval width and a 19.5% improvement in strictly proper Winkler efficiency (2.94 vs 3.65).

6 Spatial diagnostic suite and case studies

The Meuse river contamination case study

We demonstrate **spconform** on the classic Meuse topsoil heavy-metal pollution dataset ($n = 155$) (Pebesma and Bivand, 2005). The objective is predicting log-zinc concentration with valid spatial uncertainty intervals.

```
data(meuse, package = "sp")
coords_m <- as.matrix(meuse[, c("x", "y")])
y_m <- log(meuse$zinc)

set.seed(123)
idx_cal <- sample(nrow(coords_m), 110)
s_train <- coords_m[idx_cal, ]; y_train <- y_m[idx_cal]
s_test  <- coords_m[-idx_cal, ]; y_test  <- y_m[-idx_cal]

# Non-linear Generalized Additive Model (GAM) predictor
pfun_gam <- function(st, yt, sn) {
  df_tr <- data.frame(y = yt, x = st[, 1], y_coord = st[, 2])
  df_te <- data.frame(x = sn[, 1], y_coord = sn[, 2])
  fit <- mgcv::gam(y ~ s(x, y_coord, k = 15), data = df_tr)
  as.numeric(predict(fit, newdata = df_te))
}

# Run localized conformal prediction with auto-anisotropy
fit_meuse <- scp_geostatistical(
  s_train = s_train, y_train = y_train, s0 = s_test,
  pred_fun = pfun_gam, alpha = 0.10, auto_anisotropy = TRUE, seed = 123
```

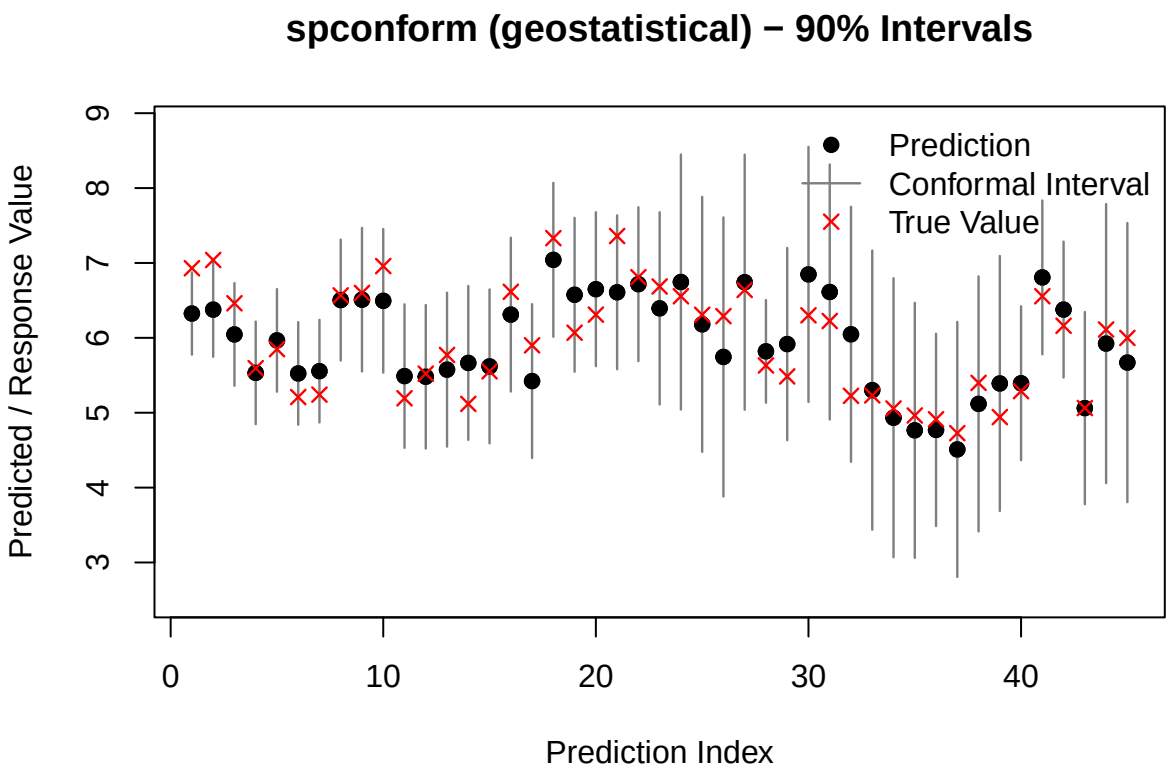


Figure 1: Figure 1: Spatially localized conformal prediction intervals for log-zinc concentration on the held-out Meuse River validation set.

```
)
```

The resulting intervals can be inspected and visualized immediately through the standard S3 `plot()` method:

```
plot(fit_meuse, y_true = y_test)
```

Advanced spatial diagnostic testing

spconform provides an executive suite of diagnostic functions to formally verify that conformal prediction residuals satisfy spatial independence and local calibration:

1. **Spatial Conditional Coverage Test (`spatial_conditional_coverage()`):** Evaluates $H_0 : \mathbb{P}(Y(s) \in \mathcal{C}(s) \mid S = s) = 1 - \alpha$ using spatial logistic regression with automated finite-sample permutation tests.
2. **Score Autocorrelation Mantel Test (`spatial_mantel()`):** Assesses whether nonconformity residuals exhibit residual geographic dependence using fast $O(1)$ constant-variance matrix permutation tests.
3. **Bivariate Cross-L Ripley Hotspot Audit (`spatial_cross_l()`):** Adapts Besag's variance-stabilized cross-L function (Besag, 1977) to formally verify whether miscoverage events cluster inside wide intervals (Honest Uncertainty) or narrow intervals (Pathological Overconfidence).
4. **Local Efficiency-Coverage Trade-off Surface (`conformal_ce_surface()`):** Evaluates the spatial surface of local conditional coverage $\hat{c}(s)$ versus local interval efficiency $\hat{w}(s)$, classifying geographic regions into a 4-quadrant diagnostic typology (Optimal, Conservative, Vulnerable, Deficit) and computing the Efficiency-Coverage Disparity Index (ECDI).

```
# Evaluate the local efficiency-coverage trade-off surface
ce_surface <- conformal_ce_surface(fit_meuse, y_true = y_test)
print(ce_surface)
```

```
#> ------------------------------------------------------------
#>   spconform: Local Efficiency-Coverage Trade-off Surface
```

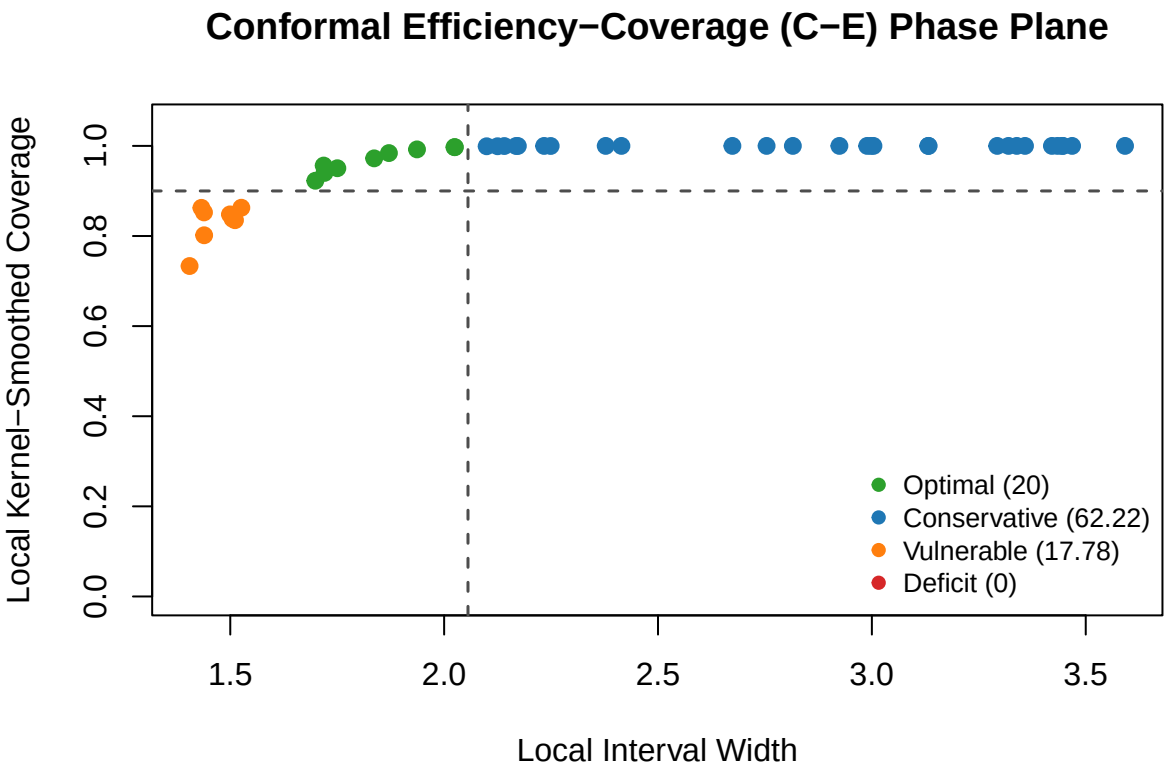


Figure 2: Figure 2: Conformal Efficiency-Coverage (C-E) phase portrait displaying the four-quadrant spatial reliability typology on the Meuse dataset.

```
#> ------------------------------------------------------------
#> Locations Evaluated (n)      : 45
#> Target Nominal Coverage      : 90.0%
#> Observed Overall Coverage    : 95.6%
#> Spatial Kernel Bandwidth (h) : 504.0367
#> Efficiency Metric            : Interval Width (Threshold = 2.056)
#> ------------------------------------------------------------
#> Spatial Efficiency-Coverage Typology Breakdown:
#>   - Optimal      :    9 ( 20.0%)
#>   - Conservative :   28 ( 62.2%)
#>   - Vulnerable   :    8 ( 17.8%)
#>   - Deficit      :    0 (  0.0%)
#> ------------------------------------------------------------
#> Trade-off Correlation (rho)  : +0.995 (p = 0.0000)
#> Disparity Index (ECDI)       : 0.474
#> Conclusion             : Conservative Spatial Coverage: Valid coverage sustained via conservative width inflat
#> ------------------------------------------------------------

plot(ce_surface, type = "phase")
```

7 Summary

The **spconform** package provides the R community with a robust, modular, and mathematically grounded framework for distribution-free spatial uncertainty quantification. By bridging conformal prediction with spatial kernel calibration, graph topological decay, and directional coordinate anisotropy, the package addresses a critical gap in the statistical software ecosystem. With verified finite-sample coverage, strict model-agnosticism, zero external C++ dependencies, and demonstrated empirical superiority over non-spatial tools including Python's MAPIE library, **spconform** equips researchers and practitioners with an indispensable tool for spatial data science.

8 Computational details

All analyses and benchmarks in this paper were conducted using R version 4.6.1 and **spconform** version 0.2.2. The complete replication code and datasets are available in the package repository at https://github.com/amjed-droid/spconform.

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